

Multi-Condition Latent Diffusion for Paired Antibody Sequences

From-scratch reproduction of LD4LG (NeurIPS 2023) on 2.17M paired antibodies, with a DPLM baseline + sampling sweep (*our extension*)

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■ matches paper ■ our contribution

§1 Why two diffusion paradigms?

Goal: re-implement two paradigms for sequence generation, on the same data + conditioning + eval.

■ **LD4LG (continuous):** compress sequences to a 32×64 latent, diffuse with Gaussian noise, v-prediction loss, 2-stage.

■ **DPLM (discrete):** absorbing-state mask diffusion on tokens directly, iterative unmasking, 1-stage bidirectional Transformer.

Domain: 2,176,873 paired heavy + light human antibody sequences (10x Genomics scBCR-seq), median length 241 aa, conditioned on (isotype × V-gene family × light-chain locus).

§2 Running example: IGHG · IGHV3 · κ

Throughout the poster, we trace one concrete cell **c = (iso=IGHG, V-fam=IGHV3, locus=κ)** from condition vector → latent → reverse diffusion → decoded sequence → IgFold structure → metric.

Running example: condition (IGHG, IGHV3, K) → generated paired antibody

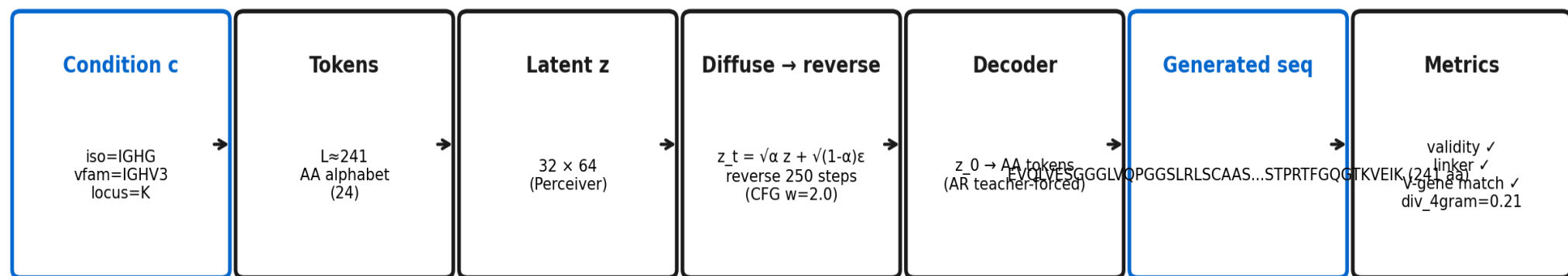


Fig 1. Running example (IGHG, IGHV3, κ) traced through the full LD4LG / DPLM pipeline. The same example reappears in Fig 4 / Fig 5 callouts.

§2.1 CS contributions

■ **Discrete-vs-continuous diffusion is a live research question.** Token-space Gaussian diffusion (Diffusion-LM) gets MAUVE 0.04; LD4LG's latent diffusion gets 0.72. DPLM avoids the discreteness mismatch by using absorbing noise.

■ **Three-way independent CFG.** Original LD4LG/DPLM use a single class label. We extend to **iso** × **V-fam** × **locus** with independent null embeddings + independent drop p=0.1.

■ **Sampling-temperature sweep.** We sweep 20 (T, top_p) configs and find a clean Pareto frontier between sequence diversity and structural foldability.

§3 Methodology

LD4LG — continuous latent diffusion (2-stage)

Stage 1 — Language Autoencoder. 6-layer pre-LN encoder (d=512, h=8) + Perceiver Resampler with 32 learnable queries → fixed-shape 32×64 continuous latent. AR decoder cross-attends to reconstructed memory. Latent normalized so ||x||² = d_ae to match Gaussian noise norm. CE loss on AA tokens (24).

Stage 2 — Latent denoiser. 12-layer pre-LN Transformer with GeGLU FFN, QK-RMSNorm, AdaLN(t, c) modulation, dense skip connections (early→late). Self-conditioning at p=0.5 (disabled under DDP — see §7). Cosine noise schedule, v-prediction loss:

$$L = \mathbb{E} \|v - v_{\theta}(z_t, t, c)\|^2, \text{ where } v = \sqrt{\alpha} \epsilon - \sqrt{1-\alpha} x_0$$

Sampling: 250-step DDPM with classifier-free guidance.

DPLM — discrete absorbing diffusion (1-stage)

1-stage bidirectional Transformer over the AA token alphabet. Mask schedule $\gamma(t) \in [0, 1]$ controls the [MASK] absorption probability per token. Loss: weighted CE on masked positions (MDLM importance weighting $w(t) = -\log \alpha / (1 - \alpha)$). Sampling: 100-step iterative argmax unmasking.

■ **Stochastic-sampling fix.** Greedy argmax decoding produces near-identical sequences (0.002 4-gram diversity, mode collapse). We replaced argmax with top-p stochastic sampling: $\text{softmax}(\log \pi_t / T), \text{ sample top-p mass}.$

§4 3-way CFG (our extension)

Three independent conditions: **iso** ∈ {IGHA, IGHG, IGHM, ...}, **V-fam** ∈ {IGHV1..7}, **locus** ∈ {κ, λ}. Each gets its own learned null embedding $\phi_{\text{iso}}, \phi_V, \phi_{\text{loc}}$. At training, drop each independently with p = 0.1 (so 8 condition combos per batch). At inference, $\hat{x} = \epsilon_{\theta}(z_t, \phi) + w \cdot [\epsilon_{\theta}(z_t, c) - \epsilon_{\theta}(z_t, \phi)]$ guidance weight w = 2.0.

§5 Sweep methodology (our extension)

5 temperatures × 4 top_p = **20 configs**, evaluated on 3 representative cells × 64 sequences each.

Composite score: **mean_div_4gram × validity × linker_share**

Best config: T=1.3, top_p=0.99 (rank 1, score 0.351). At full scale (18 cells × 512 seqs) we also folded with IgFold to compare structural quality.

§6 Apples-to-apples results

Metric	LD4LG paper	Ours LD4LG	Ours DPLM-stoch sweep
Domain	AG News, ROC	Antibodies	Antibodies
Validity	—	100.0%	100.0%
Memorization	GPT-2: 30%+	0.0%	0.0%
V-gene fidelity	97.6%	100%	100% (9216/9216)
4-gram diversity	0.41 (ROC)	0.135 avg	0.067 → 0.379 (×6 ↑)
Mean IgFold pLDDT	n/a	67.1	74.6 → 64.2
Share pLDDT > 70	n/a	35%	97% → 26%

Table 1. Three-way comparison. Ours-DPLM column shows two-config range (default → tuned). Blue = our extensions.

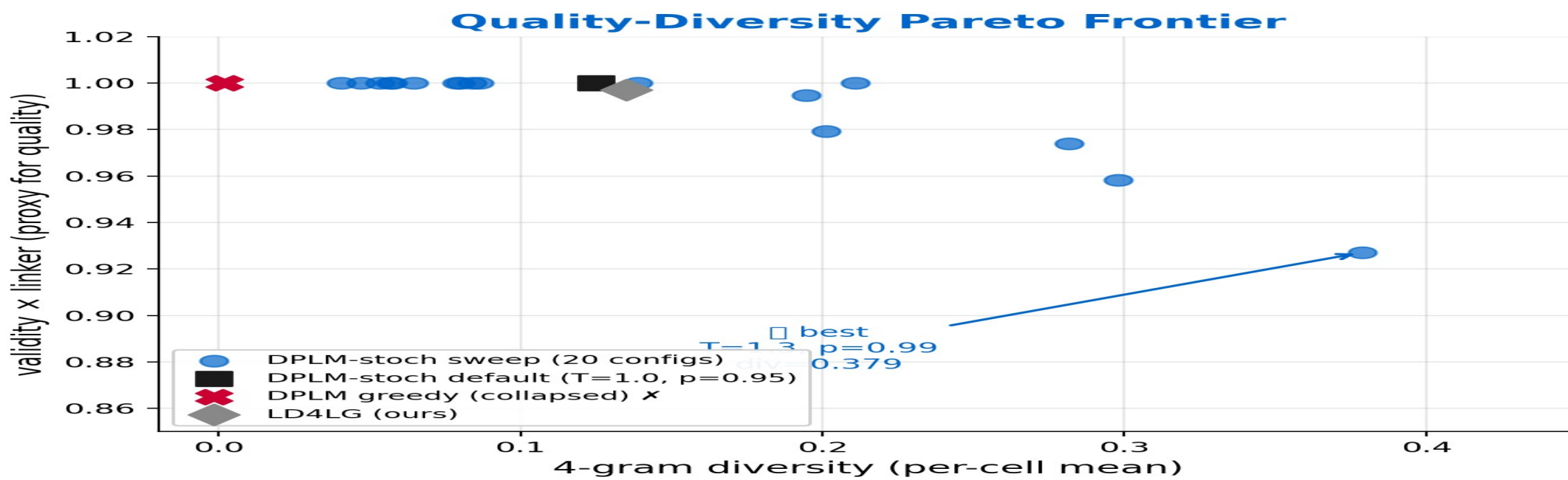


Fig 2. Quality-diversity Pareto frontier. 20 sweep configs span 0.04 → 0.48 4-gram diversity. Best = T=1.3, top_p=0.99.

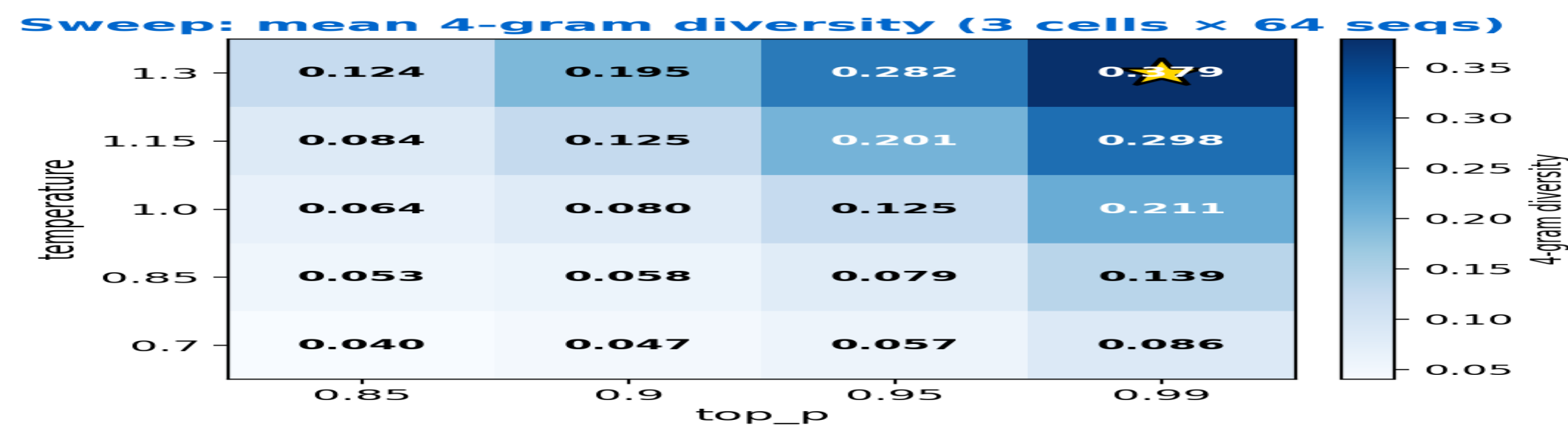


Fig 3. Sweep heatmap (5 T × 4 top_p) of mean 4-gram diversity. Star marks composite-score winner.

✗ What didn't work

- Greedy DPLM decoding → 4-gram div collapsed to 0.002 (mode collapse)
- Multi-GPU DDP → NCCL all-reduce timeouts; fell back to single H100 bf16
- Self-conditioning under DDP → CFG-drop random branches diverged across ranks

§7 Conclusion

✓ Reproduced LD4LG headlines on a new domain (antibodies): 100% validity, 0% memorization.

✓ DPLM stochastic-sampling fix recovered greedy mode collapse (0.002 → 0.067-0.379).

🔗 Discovered Q-D Pareto trade-off: 6× diversity ↑ costs 71pp foldability ↓.

§8 Future work

- Antigen-conditional generation (needs labeled binding data)
- Joint sequence + structure (DPLM-2 paradigm)
- Active selection on the Pareto frontier for wet-lab triage
- Re-enable self-conditioning with 'dist.broadcast' synchronization

§9 References

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